Question Bank
IDC Biochemistry (BCM-H-IDC)

One-word answer (x1)

1. Give an example of a secondary messenger. (Ans: PIP2, IP3, DAG, Ca++, cAMP, cGMP NO (any one))
2. Write down the breakdown product of PIP2. (Ans: Inositol triphosphate (IP3) and diacylglycerol (DAG))
3. What is the enzyme needed for cGMP production? (Ans: Guanyl cyclase)
4. What is the enzyme for degradation of PIP2? (Ans: Phospholipase C (PLC))
5. What is the secondary messenger that is activated by cAMP? (Ans: Protein Kinase A (PKA))
6. What is a simple triacylglycerol? (Ans: same fatty acids)
7. What are waxes? (Ans: alcohol other than glycerol and fatty acids.)
8. Which enzyme degrades phospholipids? (Ans: Phospholipases)
9. What is the other name for lecithin? (Ans: Phosphatidylcholine)
10. Name a minor membrane lipid which is involved in cell signalling. (Ans: Phosphatidylinositol)
11. Name the monomer subunits which form Nucleic acids. (Ans: Nucleotides)
12. Name the abundant protein in the biosphere. (Ans: RUBISCO-Ribulose bisphosphate carboxylase oxygenase)
13. Give one example of gram positive bacteria and one gram negative bacteria.
14. Name one fat soluble vitamin. (Ans. Vitamin A or D or E or K)
15. Name one water soluble vitamin ..(Ans.Vitamin C)
16. What is the other name of vitamin B1 (Ans. Thiamin)
17. What is the other name of vitamin B2 (Ans. Riboflavin)
18. What is the other name of vitamin C (Ans. Ascorbic acid)
19. Name the sugar present in DNA (Ans -Deoxyribose)
20. Name the sugar present in RNA (Ans- Ribose)
21. Name the uncommon base present in RNA (Ans -Uracil)
22. Name the type of bond between two nitrogenous bases in DNA (Ans- Hydrogen)
23. How many hydrogen bonds are present between G and C ? (Ans- 3)
24. Name an aromatic amino acid. (Tyrosine/Tryptophan/Phenylalanine)
25. Name an polar uncharged amino acid. (Serine/Threonine/Cysteine).
26. Name a basic amino acid. (Lysine/Arginine).
27. What is the approximate pKa value of the acidic group of amino acid? (2.2)
28. Name a rare amino acid. (Pyrrolysine/ Selenocysteine)
29. Write down two main functions of carbohydrates. (storage and structural framework) .

30. Write down the general formula of carbohydrates. (C(H2O)n)

Short answer (x2)

1. Why is a hormone called primary messenger?
2. Which hormones need the secondary messengers to function?
3. What is G-protein?
4. What are glycerophospholipids?
5. State one point of difference between sphingophospholipid and sphingoglycolipid.
6. Draw the structure of palmitic acid.
7. What are the functions of the cell wall in plant cells?
8. Write down the name of different classes of biomolecules.
9. What are vitamin D deficiency syndromes?
10. Name different natural sources of vitamin E.
11. What is meant by antioxidant and name one vitamin with antioxidant property
12. Why vitamin K is important for blood clotting.
13. Name the amino acids which cause fluorescence in protein.
14. Which amino acid has maximum fluorescence? What is the fluorescence maxima?
15. What is meant by an isoelectric point?
16. What is the ratio of Purine and Pyrimidine bases in DNA?
17. What is meant by melting temperature(™)?
18. What is a nucleoside?
19. What do you understand by mutarotation of glucose?
20. What is Z-DNA?
21. What are the different types of RNA?
22. Why is the monosaccharide glucose soluble in water?
23. What are vitamin C deficiency syndromes?
24. What are vitamin B12 deficiency syndromes?

SHORT QUESTION ANSWERS:

1. A hormone is called a primary messenger because it carries the signalling message from the source of synthesis to the target gland and there it binds to the receptors and stimulates the effector molecules to invoke the response.

2. The water-soluble hormones (Releasing hormones of hypothalamus, trophic hormones of anterior pituitary, posterior pituitary hormones, GI tract hormones) need the secondary messengers to function.

3. G-protein is the seven-pass transmembrane protein with GTAP-ase activity which binds to the cell-surface receptors and undergoes conformational changes when ligands bind to the GPCRs (G-protein coupled receptors).

4. Definition, structure, types, 2 functions (provided in study material in class)

5. Definition should include the point of charge , structure

6. Structure must with numbering of carbon atoms

7. Functions of cell wall are:

i) Cell wall protects the cell contents,

ii) Provides rigidity, strength and protection against mechanical stress.

iii) Allows the organism to build and hold its shape (Morphogenesis)

8. There are four major classes of Biomolecules – Carbohydrates, Proteins, Nucleic acids and Lipids.

13. Tyrosine, Tryptophan, Phenylalanine
14. TRYPTOPHAN. Fluorescence maxima is 280 nm.
15. The pH at which the net charge on amino acid is zero.
9. For younger Ricket and for older people Osteomalacia and Osteoporosis
10. Milk,meat,butter,egg,vegetable seed oil
11.Easily oxidised itself and protect other from oxidation.E.g. Vitamin E and Vitamin A
12.Vitamin K is a cofactor which help activation of enzymes responsible of blood coagulation.E.g. Prothrombin.
16. Purine : Pyrimidine (1:1)
17.Temperature at which 50 percent of DNA molecule becomes single stranded.(™)
18. A ribose sugar-a nitrogenous base.
19. Definition with pictorial representation and specific rotation value of alpha-D and beta-D glucose.
20. DNA in which the double helix has a left-handed rather than the usual right-handed twist and the sugar phosphate backbone follows a zigzag course.
21. mRNA, rRNA, tRNA.
22. Glucose, a polar organic solid, has five OH groups per molecule. It dissolves in water because of the hydrogen bonds that form between the water molecules and the OH groups.
23. Vitamin C deficiency syndromes are conditions that occur when the body has a severe lack of vitamin C (ascorbic acid). The most well-known syndrome is scurvy, which can cause a range of symptoms including:

- Fatigue and weakness
- Malaise (general feeling of illness)
- Connective tissue defects (e.g., poor wound healing, bleeding gums)
- Joint pain and swelling
- Poor immune function
- Anemia (low red blood cell count)
- Swollen limbs
- Hemorrhages (bleeding) in the skin, muscles, and internal organs

24. Deficiency syndromes can cause a range of symptoms and conditions, including:

Anemia (macrocytic anemia):
Neurological disorders:
Gastrointestinal issues:
Cardiovascular risks:
Headaches
Dizziness
Muscle weakness
Joint pain
Skin problems (acne, dermatitis)

Broad questions:

1. What is a secondary messenger? Describe with one example. (2+3)
2. Why G-protein is important? (5)
3. How PIP2 act as secondary messenger? (5)
4. What are triacylglycerols? Discuss the classification of fatty acids. (2+3)
5. What are the different types of structural lipids? Give structures. (2+3)
6. What are the major differences between prokaryotes and eukaryotes? What is Nucleoid? (4+1)
7. Globular protein dissolved in aqueous solution- what is the biochemical reason behind it. (5)
8. Fill in the blanks with the terms given: Nucleus, chromosomes, cell wall, cell membrane, protoplasm, cytoplasm, ribosome, cell organelles

The outermost layer of plant cells is the__(a)__ beneath which is the__(b)__ . The term__(c)___ refers to the jelly-like substance containing all the__(d)__ . The __(e)__ contains thread-like structures called__(f)__ .

9. Name the different sources of vitamin A? What are the different physiologic effects of vitamin A. Define coenzyme ,cofactors,prosthetic group. (2+8+5)
10. What are the different sources of vitamin D?How vitamin D is synthesized inside the human body? Describe different functions of vitamin D. Why vitamin is considered as prohormone. (2+3+8+2).
11. Name the five nitrogenous bases present in nucleic acid. What are the different components of nucleic acid? What are the major functions of nucleic acid? (3+6+6)
12. Writedown the Watson Crick structural model for DNA molecules. What are the different types of DNA molecules? (10+5)
13. What is the utility of Ramachandran plot? What is the phi and shi angle of the protein? Draw the Ramachandran plot. (4+4+7)
14. Are amino acids optically active? If so , why? Write a condensation reaction between tyrosine and alanine to form a peptide bond. Label the phi and shi angles in the peptide bond? (5+5+5)

15. Draw two acidic amino acids. Draw the pH metric titration curve for alanine titrated with NaOH. Label the isoelectric point. Why does the Peptide bond have a double bond characteristic? Explain. (4+7+4)

16. What do you understand by epimer, anomer and diastereomer?

17. Write down the Fischer projection and Haworth -Perspective formula of Galactose and Glucose.

BROAD QUESTION ANSWERS

1. Secondary messengers are the intracellular signaling molecules released by the cells in response to exposure to extracellular signaling molecules (ligands/ water-soluble hormones) that binds to the cell-surface receptors. Eg. ACTH acts through cAMP system where ACTH secreted from anterior pituitary binds to the receptors on the target glands (adrenal cortex) and stimulates G-protein which in turn stimulates adenyl cyclase that breaks down ATP to cAMP. The latter stimulates protein kinase A and the proteins are phosphorylated which then activates the DNA molecules to start transcription. (or any other example like PLC, PKC, Ca++-Calmodulin complex, PIP2-IP3+DAG etc)

2. G-protein is important because it is associated with a number of receptors on cell-surfaces where upon ligand-receptor binding stimulates the breakdown of GTP to GDP+Pi. This results in an altered conformation of different subunits of G-protein. This latter activity stimulates the downstream signaling cascade of secondary messengers (cAMP, cGMP, PKA, PKC etc). Thus, the effects/ responses can be generated in the cells through this mechanism.

3. PIP2 (phosphatidyl inositol 4, 5-biphosphate), a minor phospholipid of cell membranacts as an intermediate in IP3/DAG signaling pathway. The hormone (like adrenalin) binding to GPCR (like adrenergic receptors) stimulates PLC which catalyzes PIP2 to IP3 (inositol tri phosphate) and DAG (diacylglycerol). IP3 enters cytoplasm and stimulates Ca++ channels of SER and thereby increases the release of Ca++ which then binds to calmodulin and forms Ca-calmodulin complex and initiates the lower cascades. DAG remains on the cell membrane and stimulates PKC (protein kinase C), the latter phosphorylates a number of cytoplasmic proteins and activates them for further signaling.

4. Glycerol plus fatty acids. Structure to be given. (study material supported)

Chain length, no. of carbon atoms and no. of double bonds, each class to be mentioned with structures. (supported in detail in study material.)

5. Phospholipids : definition, glycerpophospholipid definition and structure, sphingophospholipids definition and structure, types and definitions of sphingoglycolipids. (detailed present in study material).

6.

	Prokaryotes	Eukaryotes
Type of Cell	Always unicellular	Unicellular and multi-cellular
Cell size	Ranges in size from 0.2 µm – 2.0 µm in diameter	Size ranges from 10 µm – 100 µm in diameter
Cell wall	Usually present; chemically complex in nature	When present, chemically simple in nature
Nucleus	Absent. Instead, they have a nucleoid region in the cell	Present
Ribosomes	Present. Smaller in size and spherical in shape	Present. Comparatively larger in size and linear in shape
DNA arrangement	Circular	Linear
Mitochondria	Absent	Present
Cytoplasm	Present, but cell organelles absent	Present, cell organelles present
Endoplasmic reticulum	Absent	Present
Plasmids	Present	Very rarely found in eukaryotes
Ribosome	Small ribosomes	Large ribosomes
Lysosome	Lysosomes and centrosomes are absent	Lysosomes and centrosomes are present
Cell division	Through binary fission	Through mitosis
Flagella	The flagella are smaller in size	The flagella are larger in size
Reproduction	Asexual	Both asexual and sexual
Example	Bacteria and Archaea	Plant and Animal cell

Nucleoid: The nucleoid (meaning nucleus-like) is an irregularly-shaped region within the cell of a prokaryote that contains all or most of the genetic material.

7. Globular proteins are folded such that their tertiary structure consists of the polar, or hydrophilic, amino acids arranged on the outside and the nonpolar, or hydrophobic, amino acids on the inside of the three-dimensional shape. This arrangement is responsible for the solubility of globular proteins in water.

8. a. cell membrane, b. Cell wall, c. cytoplasm, d. Cell organelles, e. Nucleus, f. Chromosomes

9.Sources of vitamin A include: green leafy vegetables, germination cereals and pulses, meat, green mango,papaya, pumpkin, yellow fruits and vegetables. Other sources are eggs, spinach, cabbage, carrots,amaranth, angle liver oils, cod liver oil, liver or halibut liver oil.

- Vision ,Resistance to infectious disease, bone remodeling, reproduction.
- Coenzymes are nonprotein organic molecules that bind loosely to an enzyme
- Cofactors are inorganic species or at least nonprotein compounds that aid enzyme function by increasing the rate of catalysis.
- Prosthetic group are tightly or covalently bound to enzyme molecule

10. Synthesised in body,milk,egg,liver,butter etc.

- Calcium and Phosphorous homeostasis,Bone remodelling,
- Regulate cell proliferation and differentiation

Dehydrocholesterol—------Cholecalciferol—----------25 Cholecalciferol—------Calcitrol

It is synthesised in liver transported from liver to other organ.so can be regarded as prohormone.

11.Adenine,Guanine,Cytosine,Thymine, Uracil

Ribose or deoxyribose sugar , a nitrogenous base,a phosphate.

Convey the hereditary information to offspring.

12. Two helical polynucleotide chains are coiled around a common axis with a right handed screw sense. The chains are antiparallel,i.e they have opposite polarity.

The sugar phosphate backbones are on the outside and the purine and pyrimidine bases lie on the inside of the helix.

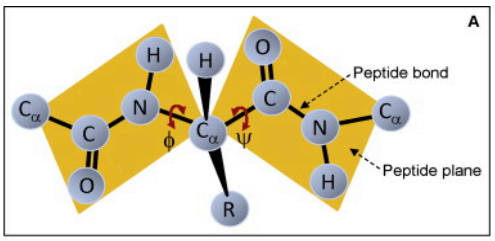
The bases are nearly perpendicular to the helix axis and adjacent bases are separated by 3.4 Å. The helical structure repeats every 34 Å and so there are 10 bases per turn of the helix.

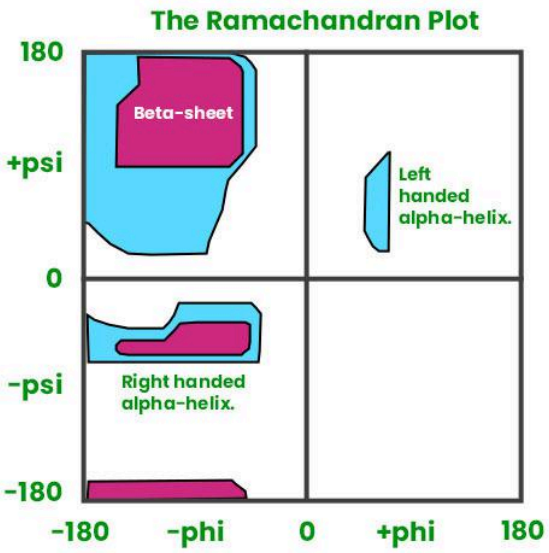
The diameter of the helix is 20 Å. This base pairs are held together by specific hydrogen bonds, which although weak but stabilize the helix because of their large number in a DNA molecule.

A DNA,B DNA,Z DNA

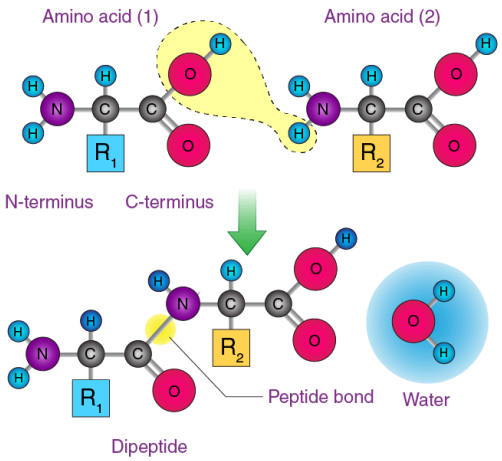
13. Ramachandran plot can be used for evaluating the accuracy of predicted protein structure.

The angle of rotation (also called torsion angle or dihedral angle) around the N Cα bond is called phi (φ) and that around the Cα C bond is called psi (ψ). These two angles largely determine the 3D shape of the polypeptide backbone of the protein.



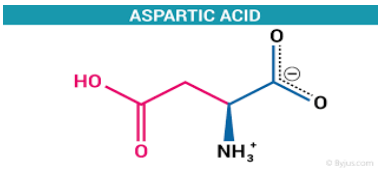


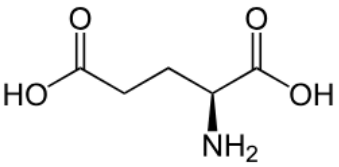
14. All amino acids are chiral in nature other than glycine because the carbon is attached to different groups, making them chiral. They can rotate the plane of polarized light.



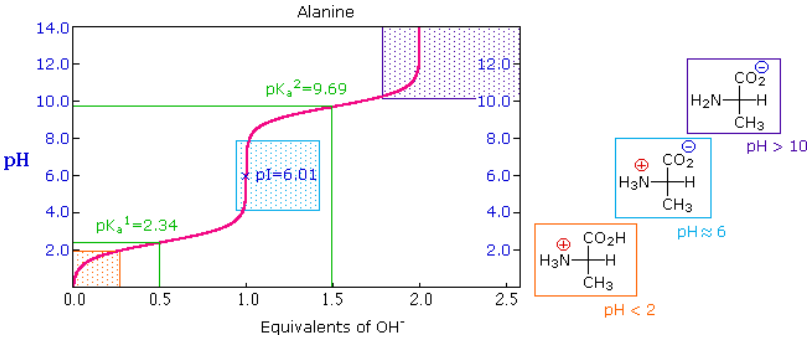
write R1 and R2 as side chains of tyrosine and alanine.

15. aspartic acid and glutamic acid

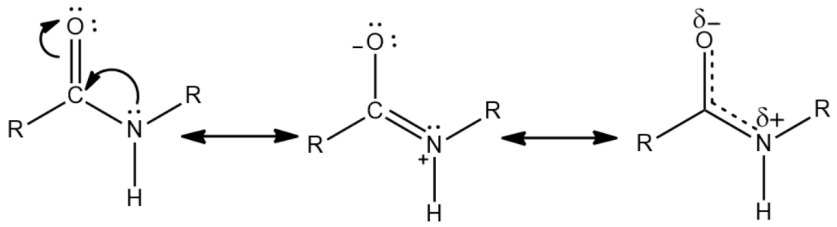




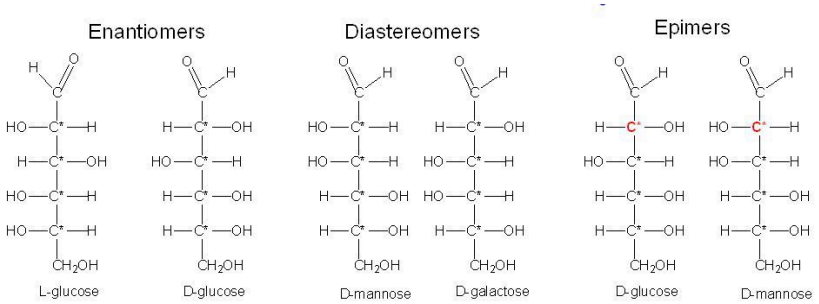
pH titration curve :



Double bond character due to resonating structures:



16. Definition with examples of monosaccharide.



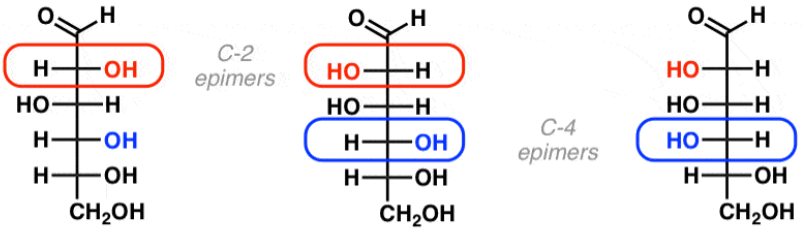
·Enantiomers = mirror images

·Pairs of isomers that have opposite configurations at one or more chiral centers but are NOT mirror images are diastereomers

·Epimers = Two sugars that differ in configuration at only one chiral center

Epimer:

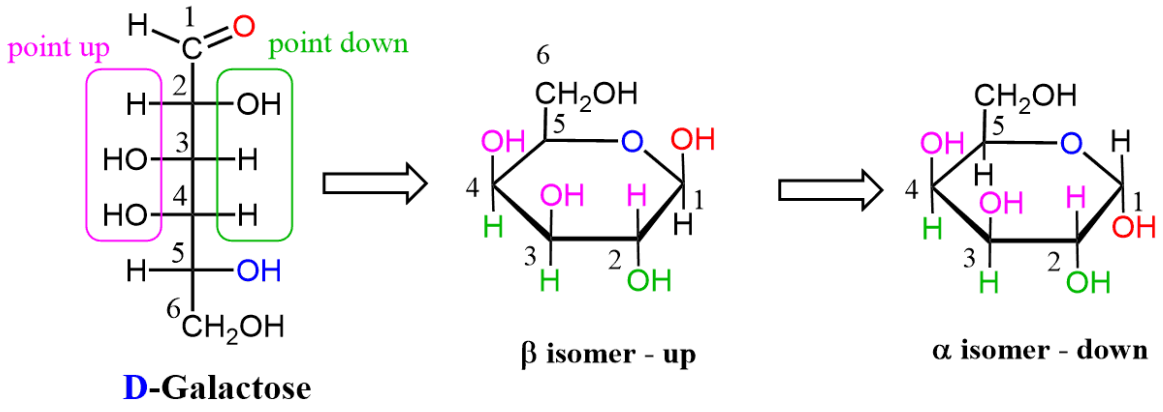
• two diastereomers which differ in their configuration only at a single chiral center

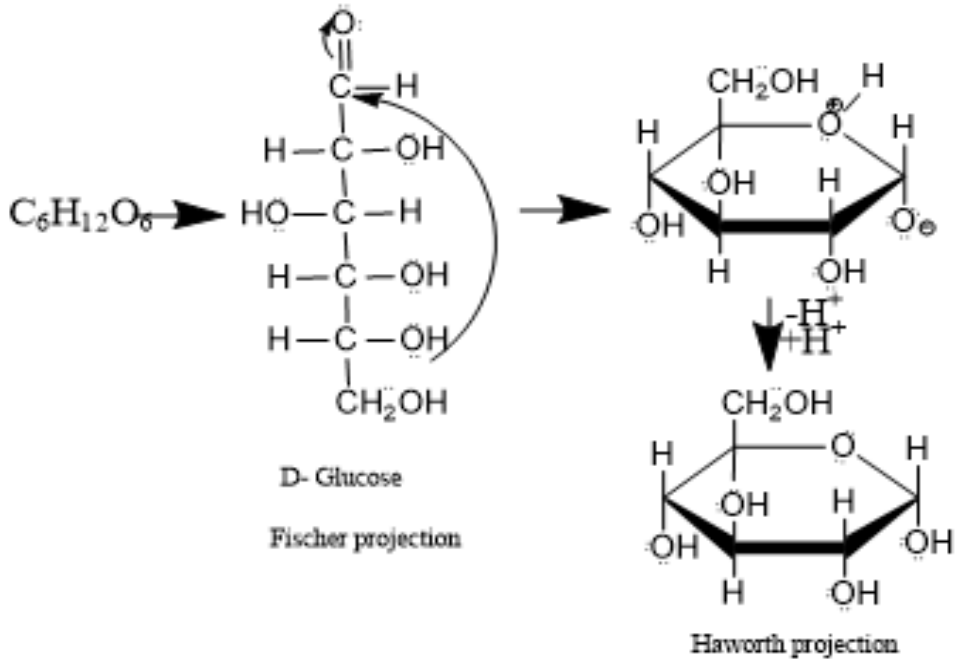


• D-Glucose and D-Mannose are epimers at C-2.
• D-Mannose and D-Talose are epimers at C-4

• D-Glucose and D-Talose are not epimers, since their configurations differ at two centers

17.




C6H12O6
D- Glucose
Fischer projection
-H+
+H+
Haworth projection